Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: medium

***Benchmark:** MMLU* | ***Context:** South Asian Agricultural and Environmental Science — Mymensingh–Telangana–Andhra Pradesh Deployment*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MMLU evaluates only text-based MCQ classification accuracy [Q23, Q35] via predicted A/B/C/D tokens [Q45]. The metric (accuracy) and calibration measures (RMS calibration error, accuracy-confidence gap) [Q60, Q62] partially align with the deployment's stated need for cross-model comparison. However, the format cannot evaluate open-ended agronomic reasoning, does not penalize language mismatches (e.g., model responding in English to a Bengali prompt), and does not support regional-language outputs. The user did not flag output form as a primary concern (MODERATE priority), and MCQ accuracy is at least usable as a baseline. But the format precludes evaluating practice-difference reasoning, generative explanations of flood-cycle decisions, or Bengali/Telugu-language extension communication — all relevant for downstream farmer/agronomist applications. Combined with the 'esoteric knowledge' data bottleneck [Q81] amplified for South Asian agricultural content, output form is partially aligned but insufficient for deployment-validity evaluation.

ID	Evidence
Q23	“we instead create a simple-to-evaluate test that measures classification accuracy on multiple choice questions.” (p.3)
Q35	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q45	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction.” (p.6)
Q60	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)
Q62	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%.” (p.7)
Q81	“Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations.” (p.8)

Strengths

- Single accuracy metric is reproducible and supports cross-model comparison [Q35, Q42]
- Calibration analysis (RMS calibration error, accuracy-confidence gap) provides useful uncertainty diagnostics for downstream deployment risk assessment [Q60, Q61, Q62, Q106]
- Per-task and per-grouping reporting [Q52, Q92] would in principle let a user isolate STEM-reasoning subscores even while ignoring US-content subscores

ID	Evidence
Q35	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q42	“All values are percentages.” (p.5)
Q52	“Figure 6 shows the accuracy of GPT-3 (few-shot) and UnifiedQA for all 57 tasks. It shows the both models are below expert-level performance for all tasks, with GPT-3's accuracy ranging from 69% for US Foreign Policy to 26% for College Chemistry.” (p.6)
Q60	“We evaluate the calibration of GPT-3 by testing how well its average confidence estimates its actual accuracy for each subject.” (p.7)

Q61	“GPT-3 is uncalibrated. In fact, its confidence is only weakly related to its actual accuracy in the zero-shot setting, with the difference between its accuracy and confidence reaching up to 24% for some subjects.” (p.7)
Q62	“Another calibration measure is the Root Mean Squared (RMS) calibration error (Hendrycks et al., 2019a; Kumar et al., 2019). Many tasks have miscalibrated predictions, such as Elementary Mathematics which has a zero-shot RMS calibration error of 19.4%.” (p.7)
Q92	“On the left are GPT-3 few shot accuracies for all of the 57 tasks. On the right are UnifiedQA transfer accuracies for all of the 57 tasks. For both models, capabilities are lopsided.” (p.11)
Q106	“While models are more calibrated in a few-shot setting than a zero-shot setting, they are still miscalibrated, with gap between accuracy and confidence reaching up to 14%. Here the correlation between confidence and accuracy is r = 0.81, compared to r = 0.63 in the zero-shot setting.” (p.13)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (single letter token) [Q45] does not match deployment needs — environmental science applications often require open-ended explanation, calculation, or recommendation. The MCQ format aligns only with the cross-model comparison goal.

OF-2: Text-to-speech is not relevant to MMLU's evaluation metric; not applicable to this benchmark form.

OF-3: Literacy considerations: Bangladesh national literacy is 77.9% [WEB-7], functional literacy 62.92% [WEB-8]; Telangana 66.54% and Andhra Pradesh 67.02% [WEB-9, WEB-11]. The downstream farmer/agronomist sub-population would not directly interact with an English-MCQ output anyway, but for any extension-tool evaluation, output form would need to support Bengali/Telugu generative output — which MMLU does not assess.

OF-4: Form mismatches are real but partial; MCQ accuracy can serve as a baseline benchmark while requiring substantial supplementation (open-ended generation, multilingual generation evaluation) for deployment-level validity.

ID	Evidence
Q35	“To measure performance on our multitask test, we compute the classification accuracy across all examples and tasks.” (p.5)
Q45	“We feed GPT-3 prompts like that shown in Figure 1a. We begin each prompt with "The following are multiple choice questions (with answers) about [subject]." For zero-shot evaluation, we append the question to the prompt. For few-shot evaluation, we add up to 5 demonstration examples with answers to the prompt before appending the question. All prompts end with "Answer: ". The model then produces probabilities for the tokens "A," "B," "C," and "D," and we treat the highest probability option as the prediction.” (p.6)
Q81	“Data may also become a bottleneck, as there is far less written about esoteric branches of knowledge than about everyday situations.” (p.8)
WEB-7	Bangladesh literacy 77.9% (BBS 2024) (https://www.observerbd.com/news/542998)
WEB-8	Bangladesh functional literacy 62.92% (BBS LAS 2023) (https://en.prothomalo.com/bangladesh/c7axno0vhh)
WEB-9	Telangana literacy 66.54% (Census 2011) (https://testbook.com/question-answer/what-is-the-literacy-rate-of-telangana--5efeb788fbc9b00d102b5fda)
WEB-11	Andhra Pradesh literacy 67.02% (Census 2011) (https://www.census2011.co.in/census/state/andhra+pradesh.html)

Information Gaps

- Whether deployment will require open-ended generation or remain MCQ-style was not fully specified in elicitation

Remediation

Gap 1: MCQ accuracy cannot capture open-ended agronomic reasoning needed for downstream farmer/agronomist applications, and does not penalize language-mismatch failures.

Recommendation: Supplement MMLU MCQ accuracy with open-ended generative evaluation in Bengali and Telugu, evaluated by regional experts using rubrics that explicitly penalize language drift (English response to Bengali prompt) and reward dialect-appropriate register for Mymensingh-Bengali agricultural extension communication.